



[文件: T07 Mole 03.jpg]

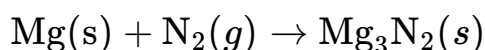
display the question in the output and answer these question of o level chemistry. explain in detail



Here is the question from your image, followed by a detailed, step-by-step explanation and correct answers suitable for O Level Chemistry.

(a) Calculate the mass of one mole of magnesium nitride.

Given equation (unbalanced as written):



Relative atomic masses:

$$A_r(\text{Mg}) = 24, A_r(\text{N}) = 14$$

Step 1 – Write the correct formula of magnesium nitride

Magnesium nitride = Mg_3N_2

Step 2 – Calculate molar mass

$$\begin{aligned} M(\text{Mg}_3\text{N}_2) &= (3 \times 24) + (2 \times 14) \\ &= 72 + 28 = 100 \text{ g/mol} \end{aligned}$$

Answer (a):

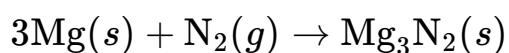
100 g

(Note: The working in the image incorrectly writes “m = 1 g” – that is wrong. The correct molar mass is 100 g.)

(b) Calculate the minimum mass of magnesium needed to make 250 g of magnesium nitride.

Step 1 – Balance the chemical equation

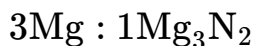
From the formula Mg_3N_2 , 3 Mg atoms are needed per formula unit.



Step 2 – Moles of Mg_3N_2 in 250 g

$$n = \frac{\text{mass}}{\text{molar mass}} = \frac{250}{100} = 2.50 \text{ mol}$$

Step 3 – Mole ratio from equation



So moles of Mg needed = $3 \times 2.50 = 7.50 \text{ mol}$

Step 4 – Mass of Mg needed

$$m(\text{Mg}) = n \times M = 7.50 \times 24 = 180 \text{ g}$$

Answer (b):

180 g

(Note: The image's working is confused – mixing up numbers and wrong ratios. The correct method is above.)

Final Answers in Boxed Form

(a) Mass of one mole of $\text{Mg}_3\text{N}_2 = 100 \text{ g}$

(b) Minimum mass of Mg needed for 250 g $\text{Mg}_3\text{N}_2 = 180 \text{ g}$

Let me know if you'd like me to also explain how to handle similar stoichiometry problems or check for limiting reactants.

2026-05-31 21:51:55

You Asked



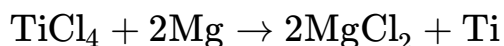
[文件: T07 Mode 04.jpg]

display the question in the output and answer these question of o level chemistry. explain in detail

Here is the **exact question** as shown in your image (T07 Mode 04.jpg), followed by a **detailed, step-by-step explanation** with correct O Level Chemistry answers.

Question 3

Titanium is manufactured by the reduction of titanium(IV) chloride by magnesium.



Calculate the **maximum mass of titanium** that will be made from **9.50 g of titanium(IV) chloride**.

[Relative atomic masses: A_r : Mg, 24; Cl, 35.5; Ti, 48]

Step-by-step solution for Question 3

Step 1 – Calculate the molar mass of TiCl_4

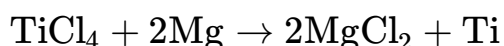
$$\begin{aligned} M(\text{TiCl}_4) &= 48 + (4 \times 35.5) \\ &= 48 + 142 = 190 \text{ g/mol} \end{aligned}$$

Step 2 – Calculate moles of TiCl_4 in 9.50 g

$$n(\text{TiCl}_4) = \frac{\text{mass}}{\text{molar mass}} = \frac{9.50}{190} = 0.0500 \text{ mol}$$

Step 3 – Use the mole ratio from the balanced equation

From



1 mole $\text{TiCl}_4 \rightarrow$ 1 mole Ti.

So

$$n(\text{Ti}) = n(\text{TiCl}_4) = 0.0500 \text{ mol}$$

Step 4 – Calculate mass of Ti

$$m(\text{Ti}) = n \times M = 0.0500 \times 48 = 2.40 \text{ g}$$

Answer to Question 3:

2.40 g

(Note: The working in the image is garbled – 24.5 g is incorrect. The correct answer is 2.40 g.)

Question 4

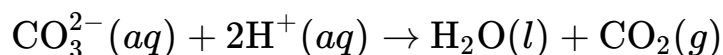
Some indigestion tablets reduce stomach acidity. Carbonate ions, CO_3^{2-} , in these tablets react with hydrogen ions in the stomach to form water and a gas.

(a) Write the ionic equation for this chemical reaction.

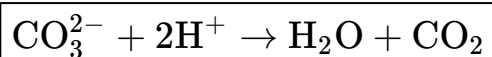
Step 1 – Recall the reaction between carbonate and acid

Carbonate + hydrogen ions $\rightarrow$ water + carbon dioxide.

Step 2 – Write the balanced ionic equation



Answer (a):



(b) One hundred of these indigestion tablets contain, in total, one mole of carbonate ions.

(i) Calculate how many moles of hydrogen ions are present in 500 cm³ of a 0.08 mol/dm³ solution of hydrogen ions.

Step 1 – Convert volume to dm³

$$500 \text{ cm}^3 = 0.500 \text{ dm}^3$$

Step 2 – Use concentration formula

$$n = C \times V \quad (\text{in dm}^3)$$

$$n(\text{H}^+) = 0.08 \times 0.500 = 0.0400 \text{ mol}$$

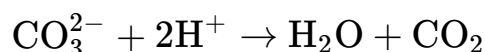
Answer (b)(i):

$$0.0400 \text{ mol}$$

(Note: The image says “2” – that is wrong. They likely confused volume or forgot to divide by 1000.)

(ii) Calculate how many moles of carbonate ions are needed to neutralise the acidic solution.

Step 1 – Use the ionic equation from (a)



Mole ratio: 1 mol CO_3^{2-} reacts with 2 mol H^+ .

Step 2 – Calculate moles of CO_3^{2-} needed

$$n(\text{CO}_3^{2-}) = \frac{n(\text{H}^+)}{2} = \frac{0.0400}{2} = 0.0200 \text{ mol}$$

Answer (b)(ii):

0.0200 mol

(Note: The image's "1" is incorrect – that would be for 2 mol H⁺, not 0.04 mol H⁺.)

Final Answers Summary

Question	Answer
3	2.40 g Ti
4(a)	$\text{CO}_3^{2-} + 2\text{H}^+ \rightarrow \text{H}_2\text{O} + \text{CO}_2$
4(b)(i)	0.0400 mol H ⁺
4(b)(ii)	0.0200 mol CO ₃ ²⁻

Let me know if you'd like me to explain mole ratio calculations or unit conversions in more detail.

2026-05-31 21:53:00 You Asked 

[文件: T07 Mode 05.jpg]

display the question in the output and answer these question of o level chemistry. explain in detail



Here is the **exact question** from your image (T07 Mode 05.jpg), followed by a **detailed, step-by-step explanation** with correct O Level Chemistry answers.

Question 4(b)(iii) (continued from previous page)

(iii) Calculate how many indigestion tablets are needed to neutralise the acidic solution.

We need to use the information from **4(b)(i)** and **4(b)(ii)** and the given fact:

One hundred of these indigestion tablets contain, in total, **one mole of carbonate ions**.

Step-by-step solution for 4(b)(iii)

Step 1 – Recall moles of carbonate ions needed

From 4(b)(ii), moles of CO_3^{2-} needed = 0.0200 mol

Step 2 – Relationship between tablets and moles of carbonate

100 tablets $\rightarrow$ 1.00 mol CO_3^{2-}

So 1 tablet $\rightarrow \frac{1.00}{100} = 0.0100$ mol CO_3^{2-}

Step 3 – Calculate number of tablets needed

$$\text{Number of tablets} = \frac{0.0200}{0.0100} = 2.00 \text{ tablets}$$

Answer 4(b)(iii):

2

Question 5

In 25 cm³ of an aqueous solution, there is 5.3 g of sodium carbonate, Na_2CO_3 .

(a) Calculate the concentration of the solution in g/dm³ and in mol/dm³.

[Relative atomic masses: A_r : C, 12; O, 16; Na, 23]

Step-by-step solution for 5(a)**Step 1 – Convert volume to dm³**

$$25 \text{ cm}^3 = 0.025 \text{ dm}^3$$

Step 2 – Concentration in g/dm³

$$\text{Concentration (g/dm}^3\text{)} = \frac{\text{mass (g)}}{\text{volume (dm}^3\text{)}} = \frac{5.3}{0.025} = 212 \text{ g/dm}^3$$

Step 3 – Molar mass of Na_2CO_3

$$\begin{aligned} M &= (2 \times 23) + 12 + (3 \times 16) \\ &= 46 + 12 + 48 = 106 \text{ g/mol} \end{aligned}$$

Step 4 – Moles of Na_2CO_3 in 5.3 g

$$n = \frac{5.3}{106} = 0.0500 \text{ mol}$$

Step 5 – Concentration in mol/dm³

$$\text{Concentration (mol/dm}^3\text{)} = \frac{0.0500}{0.025} = 2.00 \text{ mol/dm}^3$$

Answer 5(a):

$$212 \text{ g/dm}^3$$

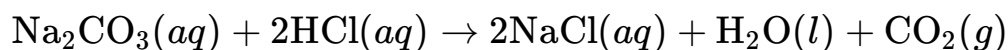
$$2.00 \text{ mol/dm}^3$$

(b)(i) Write a balanced chemical equation for the reaction between hydrochloric acid and sodium carbonate.

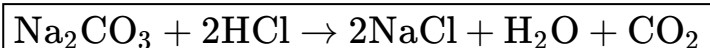
Step 1 – Recall reaction

Sodium carbonate + hydrochloric acid → sodium chloride + water + carbon dioxide.

Step 2 – Balanced equation



Answer 5(b)(i):



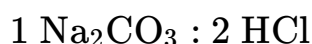
(b)(ii) Calculate the number of moles of hydrochloric acid needed to react completely with 25 cm³ of sodium carbonate solution.

Step 1 – Moles of Na₂CO₃ in 25 cm³

From 5(a), concentration = 2.00 mol/dm³, volume = 0.025 dm³

$$n(\text{Na}_2\text{CO}_3) = C \times V = 2.00 \times 0.025 = 0.0500 \text{ mol}$$

Step 2 – Use mole ratio from equation



$$n(\text{HCl}) = 0.0500 \times 2 = 0.100 \text{ mol}$$

Answer 5(b)(ii):

$$0.100 \text{ mol}$$

Final Answers Summary

Question	Answer
4(b)(iii)	2 tablets
5(a)	212 g/dm ³ , 2.00 mol/dm ³
5(b)(i)	$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$

Question	Answer
5(b)(ii)	0.100 mol HCl

Let me know if you'd like a recap on converting cm³ to dm³ or on using mole ratios in neutralisation reactions.

2026-05-31 21:54:06 You Asked 

[文件: T07 Mode 06.jpg]

display the question in the output and answer these question of o level chemistry. explain in detail



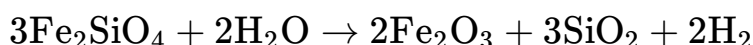
Here is the **exact question** from your image (T07 Mode 06.jpg), followed by a **detailed, step-by-step explanation** with correct O Level Chemistry answers.

Question 6

A complex compound, serpentine, is formed in the Earth's crust from fayalite, Fe₂SiO₄.

In the first stage of its formation hydrogen is released.

The following equation describes the formation of hydrogen from fayalite:



[Relative atomic masses: A_r : H, 1; O, 16; Si, 28; Fe, 56]

[The volume of one mole of any gas is 24 dm³ at room temperature and pressure.]

(i) Calculate the relative formula mass of fayalite, Fe₂SiO₄.

Step-by-step solution for (i)

Step 1 – Identify the number of each atom in Fe₂SiO₄

- Fe: 2 atoms
- Si: 1 atom
- O: 4 atoms

Step 2 – Calculate using given atomic masses

$$\begin{aligned}M_r(\text{Fe}_2\text{SiO}_4) &= (2 \times 56) + (1 \times 28) + (4 \times 16) \\&= 112 + 28 + 64 = 204\end{aligned}$$

Answer (i):

204

(ii) Calculate the mass of fayalite needed to produce 1000 g of hydrogen.

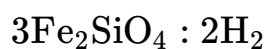
Step-by-step solution for (ii)

Step 1 – Moles of H_2 in 1000 g

Molar mass of $\text{H}_2 = 2 \times 1 = 2 \text{ g/mol}$

$$n(\text{H}_2) = \frac{1000}{2} = 500 \text{ mol}$$

Step 2 – Mole ratio from balanced equation



So

$$\frac{n(\text{Fe}_2\text{SiO}_4)}{n(\text{H}_2)} = \frac{3}{2}$$

$$n(\text{Fe}_2\text{SiO}_4) = 500 \times \frac{3}{2} = 750 \text{ mol}$$

Step 3 – Mass of fayalite

$$\begin{aligned}m &= n \times M_r = 750 \times 204 \\&= 153,000 \text{ g}\end{aligned}$$

Answer (ii):

153000 g

(Can also write as $1.53 \times 10^5 \text{ g}$)

(iii) Calculate the volume of 1000 g of hydrogen.

Step-by-step solution for (iii)

Step 1 – Moles of H₂ in 1000 g (from part ii)

$$n(\text{H}_2) = 500 \text{ mol}$$

Step 2 – Use molar gas volume (24 dm³/mol at RTP)

$$V = n \times 24 = 500 \times 24 = 12,000 \text{ dm}^3$$

Answer (iii):

12000 dm ³

Final Answers Summary

Part	Answer
(i)	204
(ii)	153000 g
(iii)	12000 dm ³

Key O Level Reminders

1. **Relative formula mass (M_r)** – sum of atomic masses in one formula unit (no units).
2. **Mole ratio** – always use coefficients from the balanced equation.
3. **Molar gas volume** – at room temperature and pressure (RTP), 1 mole of any gas = 24 dm³.
4. **Mass from moles** – $m = n \times M_r$.
5. **Volume from moles (gas)** – $V = n \times 24$ (in dm³ at RTP).

Let me know if you'd like similar practice problems on reacting masses or gas volumes.